

Topic 2: The Exponential Response Formula

Prepared by Staples Education

Topics Overview

This topic delivers a fast, systematic shortcut for finding particular solutions to constant-coefficient linear ODEs driven by exponential and sinusoidal forcing. The core sub-topics are:

- **The Core Concept** — using the characteristic polynomial $p(r)$ as an evaluation machine for the operator $p(D)$.
- **The Standard ERF** — $y_p = \frac{Ae^{at}}{p(a)}$ for forcing $g(t) = Ae^{at}$ when $p(a) \neq 0$.
- **The Modified ERF (Resonance)** — handling $p(a) = 0$ for single roots, double roots, and general multiplicity.
- **Complex Exponentials** — solving $\cos(\omega t)$ and $\sin(\omega t)$ forcing via Euler's formula and real/imaginary parts.

Part I — Condensed Cheat Sheet

The Core Concept

Consider a linear constant-coefficient operator written in terms of $D = \frac{d}{dt}$:

$$p(D)y = (a_n D^n + a_{n-1} D^{n-1} + \cdots + a_1 D + a_0)y = g(t).$$

The single fact that powers everything in this topic is that e^{at} is an *eigenfunction* of D — differentiating it simply multiplies by a . Applying the whole operator therefore reduces to a single polynomial evaluation:

$$D e^{at} = a e^{at}, \quad D^k e^{at} = a^k e^{at} \implies \boxed{p(D) e^{at} = p(a) e^{at}}.$$

This means we never have to differentiate a guess and match coefficients — the operator collapses into the number $p(a)$. Everything below is a consequence of this identity.

The Standard ERF

When the forcing is a clean exponential and a is *not* a root of the characteristic equation, the particular solution drops out immediately.

Standard Exponential Response Formula

For $p(D)y = A e^{at}$ with $p(a) \neq 0$:

$$y_p = \frac{A e^{at}}{p(a)}.$$

Procedure — write down $p(r)$, evaluate at $r = a$, divide. If $p(a) \neq 0$ you are done.

Worked Example 1 — Standard ERF

Solve for a particular solution of $y'' + 3y' + 2y = 5e^{3t}$.

Step 1 — Characteristic polynomial. $p(r) = r^2 + 3r + 2$.

Step 2 — Evaluate at the exponent $a = 3$. $p(3) = 9 + 9 + 2 = 20$.

Since $p(3) = 20 \neq 0$, the standard formula applies directly.

Step 3 — Apply the ERF.

$$y_p = \frac{5e^{3t}}{20} = \frac{e^{3t}}{4}.$$

The Modified ERF — Resonance

When a is a root of the characteristic equation, $p(a) = 0$ and the standard formula divides by zero — this is **resonance**. The forcing aligns with a natural mode of the system, so the response grows in t . The fix is to climb the derivatives of p until we hit a nonzero value, multiplying by a matching power of t .

Generalized / Modified ERF

If a is a root of $p(r)$ of multiplicity s — so that $p(a) = p'(a) = \dots = p^{(s-1)}(a) = 0$ but $p^{(s)}(a) \neq 0$ — then for $p(D)y = Ae^{at}$:

$$y_p = \frac{At^s e^{at}}{p^{(s)}(a)}.$$

Single root ($s = 1$): $y_p = \frac{At e^{at}}{p'(a)}.$ **Double root ($s = 2$):** $y_p = \frac{At^2 e^{at}}{p''(a)}.$

Worked Example 2 — Single-Root Resonance

Solve for a particular solution of $y'' - 3y' + 2y = 4e^t$.

Step 1 — Characteristic polynomial and roots. $p(r) = r^2 - 3r + 2 = (r - 1)(r - 2)$, with roots $r = 1, 2$.

Step 2 — Check the exponent $a = 1$. $p(1) = 1 - 3 + 2 = 0$ — resonance with a *single* root.

Step 3 — Use the derivative. $p'(r) = 2r - 3$, so $p'(1) = 2 - 3 = -1$.

Step 4 — Apply the modified ERF.

$$y_p = \frac{4te^t}{-1} = -4te^t.$$

Worked Example 3 — Double-Root Resonance

Solve for a particular solution of $y'' - 4y' + 4y = 6e^{2t}$.

Step 1 — Characteristic polynomial. $p(r) = r^2 - 4r + 4 = (r - 2)^2$ — a *double* root at $r = 2$.

Step 2 — Confirm the multiplicity. $p(2) = 0$ and $p'(r) = 2r - 4$ gives $p'(2) = 0$ — both vanish, so $s = 2$.

Step 3 — Second derivative. $p''(r) = 2$, so $p''(2) = 2$.

Step 4 — Apply the modified ERF.

$$y_p = \frac{6t^2 e^{2t}}{2} = 3t^2 e^{2t}.$$

Complex Exponentials — Sinusoidal Forcing

Sinusoidal forcing is handled by routing it through a complex exponential. Euler's formula $e^{i\omega t} = \cos \omega t + i \sin \omega t$ lets us treat $\cos$ and $\sin$ as the real and imaginary parts of a single exponential — then the ERF machinery applies unchanged.

Complex ERF for Sinusoidal Forcing

To solve $p(D)y = A \cos(\omega t)$ or $A \sin(\omega t)$, complexify the forcing to $A e^{i\omega t}$ and solve

$$\tilde{y}_p = \frac{A e^{i\omega t}}{p(i\omega)} \quad (\text{if } p(i\omega) \neq 0).$$

Then recover the real solution by taking the matching part:

$$g(t) = A \cos \omega t \Rightarrow y_p = \operatorname{Re}(\tilde{y}_p), \quad g(t) = A \sin \omega t \Rightarrow y_p = \operatorname{Im}(\tilde{y}_p).$$

If $p(i\omega) = 0$ the system is at resonance — use the modified ERF with $a = i\omega$ before taking parts.

Worked Example 4 — Cosine Forcing

Solve for a particular solution of $y'' + 4y = \cos(t)$.

Step 1 — Complexify. Replace $\cos t$ with e^{it} and plan to take the real part. Here $\omega = 1$, $A = 1$.

Step 2 — Evaluate p at $i\omega$. $p(r) = r^2 + 4$, so $p(i) = i^2 + 4 = -1 + 4 = 3$.

Since $p(i) = 3 \neq 0$, no resonance.

Step 3 — Complex ERF. $\tilde{y}_p = \frac{e^{it}}{3}$.

Step 4 — Take the real part. $y_p = \operatorname{Re}\left(\frac{e^{it}}{3}\right) = \frac{\cos t}{3}$.

Part II — Practice Questions

For each of the following, find a particular solution y_p using the Exponential Response Formula. Identify the appropriate case — standard, resonance, or complex — before computing.

1. $y'' + 3y' + 2y = 5e^{3t}$
2. $y'' - y = 4e^{2t}$
3. $y''' + y'' + y' + y = 2e^t$
4. $y'' - 3y' + 2y = 4e^t$
5. $y'' + y' - 6y = 7e^{2t}$
6. $y'' - 4y' + 4y = 6e^{2t}$
7. $y'' + 4y = \cos(t)$
8. $y'' + 3y' + 2y = \sin(t)$
9. $y'' + 2y' + 5y = 4\cos(3t)$

10. $y'' + 4y = \cos(2t)$

Part III — Fully Worked Solutions

Solution 1

Case: Standard ERF. The forcing is Ae^{at} with $A = 5$, $a = 3$.

$$p(r) = r^2 + 3r + 2, \quad p(3) = 9 + 9 + 2 = 20 \neq 0.$$

$$y_p = \frac{5e^{3t}}{20} = \frac{e^{3t}}{4}.$$

Solution 2

Case: Standard ERF. Here $A = 4$, $a = 2$.

$$p(r) = r^2 - 1, \quad p(2) = 4 - 1 = 3 \neq 0.$$

$$y_p = \frac{4e^{2t}}{3}.$$

Solution 3

Case: Standard ERF (third order). Here $A = 2$, $a = 1$.

$$p(r) = r^3 + r^2 + r + 1, \quad p(1) = 1 + 1 + 1 + 1 = 4 \neq 0.$$

$$y_p = \frac{2e^t}{4} = \frac{e^t}{2}.$$

Solution 4

Case: Single-Root Resonance. The forcing exponent $a = 1$ is a root.

$$p(r) = r^2 - 3r + 2 = (r - 1)(r - 2), \quad p(1) = 0.$$

The root is simple, so differentiate once:

$$p'(r) = 2r - 3, \quad p'(1) = -1 \neq 0.$$

$$y_p = \frac{4te^t}{-1} = -4te^t.$$

Solution 5

Case: Single-Root Resonance. The exponent $a = 2$ is a root.

$$p(r) = r^2 + r - 6 = (r + 3)(r - 2), \quad p(2) = 4 + 2 - 6 = 0.$$

$$p'(r) = 2r + 1, \quad p'(2) = 5 \neq 0.$$

$$y_p = \frac{7te^{2t}}{5}.$$

Solution 6

Case: Double-Root Resonance. The exponent $a = 2$ is a repeated root.

$$p(r) = r^2 - 4r + 4 = (r - 2)^2, \quad p(2) = 0, \quad p'(2) = 2(2) - 4 = 0.$$

Both p and p' vanish, confirming multiplicity $s = 2$. Use the second derivative:

$$p''(r) = 2, \quad p''(2) = 2.$$

$$y_p = \frac{6t^2 e^{2t}}{2} = 3t^2 e^{2t}.$$

Solution 7

Case: Complex Exponential — cosine forcing. Complexify $\cos t \rightarrow e^{it}$ and take the real part. Here $\omega = 1$.

$$p(r) = r^2 + 4, \quad p(i) = -1 + 4 = 3 \neq 0.$$

$$\tilde{y}_p = \frac{e^{it}}{3}, \quad y_p = \operatorname{Re}\left(\frac{e^{it}}{3}\right) = \frac{\cos t}{3}.$$

Solution 8

Case: Complex Exponential — sine forcing. Complexify $\sin t \rightarrow e^{it}$ and take the imaginary part. Here $\omega = 1$.

$$p(r) = r^2 + 3r + 2, \quad p(i) = -1 + 3i + 2 = 1 + 3i \neq 0.$$

Form the complex solution and rationalize the denominator:

$$\tilde{y}_p = \frac{e^{it}}{1 + 3i} = \frac{(1 - 3i)e^{it}}{(1 + 3i)(1 - 3i)} = \frac{(1 - 3i)e^{it}}{10}.$$

Expand using $e^{it} = \cos t + i \sin t$:

$$\tilde{y}_p = \frac{1}{10} [(\cos t + 3 \sin t) + i(\sin t - 3 \cos t)].$$

Take the imaginary part for sine forcing:

$$y_p = \operatorname{Im}(\tilde{y}_p) = \frac{1}{10} (\sin t - 3 \cos t).$$

Solution 9

Case: Complex Exponential — cosine forcing with damping. Complexify $4 \cos 3t \rightarrow 4e^{3it}$ and take the real part. Here $\omega = 3$, $A = 4$.

$$p(r) = r^2 + 2r + 5, \quad p(3i) = -9 + 6i + 5 = -4 + 6i \neq 0.$$

Rationalize:

$$\tilde{y}_p = \frac{4e^{3it}}{-4 + 6i} = \frac{4(-4 - 6i)e^{3it}}{(-4)^2 + 6^2} = \frac{(-4 - 6i)e^{3it}}{13}.$$

Expand with $e^{3it} = \cos 3t + i \sin 3t$:

$$\tilde{y}_p = \frac{1}{13} [(-4 \cos 3t + 6 \sin 3t) + i(-6 \cos 3t - 4 \sin 3t)].$$

Take the real part for cosine forcing:

$$y_p = \operatorname{Re}(\tilde{y}_p) = \frac{1}{13} (6 \sin 3t - 4 \cos 3t).$$

Solution 10

Case: Complex Resonance. Complexify $\cos 2t \rightarrow e^{2it}$ with $\omega = 2$. Test the denominator first:

$$p(r) = r^2 + 4, \quad p(2i) = (2i)^2 + 4 = -4 + 4 = 0.$$

Since $p(2i) = 0$, this is resonance — the root $a = 2i$ is simple, so use the modified ERF with the first derivative:

$$p'(r) = 2r, \quad p'(2i) = 4i \neq 0.$$
$$\tilde{y}_p = \frac{t e^{2it}}{p'(2i)} = \frac{t e^{2it}}{4i} = \frac{-i}{4} t e^{2it}.$$

Expand using $e^{2it} = \cos 2t + i \sin 2t$:

$$\tilde{y}_p = \frac{t}{4} (-i)(\cos 2t + i \sin 2t) = \frac{t}{4} (\sin 2t - i \cos 2t).$$

Take the real part for cosine forcing:

$$y_p = \operatorname{Re}(\tilde{y}_p) = \frac{t \sin 2t}{4}.$$

This grows linearly in t — the signature of an undamped system driven at its natural frequency.